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Regular Article

An assessment of graduates skills gap for sustainable employability during the 4IR in Bangladesh

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ABSTRACT

The Fourth Industrial Revolution (4IR) leads global transformation through technological advancements, streamlining labor processes, and boosting productivity. This study explores the emerging employment sectors in Bangladesh for graduates, such as data science, artificial intelligence, and e-commerce management, in the context of the 4IR. This study also examines skills discrepancies, particularly in technical and soft skills. Using a mixed-methods approach, combining quantitative surveys (n = 1320) and qualitative interviews (n = 32), data were collected from university graduates and stakeholders across Bangladesh. Findings reveal substantial skills mismatches, particularly in advanced technology and digital competencies, hindering graduates' sustainable employability. This study underlines the need for tailored curricula, industry-academia partnerships, and upskilling programs to bridge these gaps for graduate's employability. These actionable counsels directly align educational institutions with market demands (industry-academia collaboration), fostering sustainable employability in the 4IR era of Bangladesh.

1. Introduction

The Fourth Industrial Revolution (4IR) drives global transformation via technological innovation, enhancing productivity and streamlining work processes. In Bangladesh, technological advancements present an opportunity to accelerate economic development, as emphasized by Alam, Ogawa, and Ahsan (2024a, 2024b), Antoniuk et al. (2017), and Bhuiyan et al. (2020). However, they also pose challenges, such as the risk of job displacement through automation and a growing skills mismatch in the labor market (Berg et al., 2016; Bertani et al., 2020; Wright & Schultz, 2018). Artificial intelligence (AI) has the potential for increased efficiency, but it has also created an urgent need for advanced skills in data science and machine learning (Peters, 2019; Pol & Reveley, 2017).

To harness the potential of the 4IR, comprehensive skill development programs tailored to the needs of Bangladesh's young population are essential (Babu, 2021; Khatun & Saadat, 2020; Shahriar et al., 2021). The labor market faces dual challenges: a scarcity of skilled professionals in emerging sectors and limited alignment between graduates' skills and employers' demands. Sectors like IT services have seen notable expansion (Fink & Elisabetta, 2019). Yet, a significant portion of the workforce lacks the technical and soft skills necessary for these fields (Rahman & Abedin, 2021). For example, a recent study found that despite technological advances, corporations often fail to invest sufficient resources in employee retraining and upskilling (Mujeri & Mujeri, 2020).

While numerous studies explore sustainable employment (Fleuren et al., 2016; Hazelzet et al., 2019; Basharat & Alam, 2024) and the impacts of the 4IR in Bangladesh (Latif et al., 2021; Rahman & Taher, 2022), limited research specifically examines emerging sustainable employability areas and the graduates' skills gap during the 4IR. For instance, there is little focus on how Bangladesh can prepare its workforce for AI engineering, digital marketing, and cybersecurity. This study addresses these gaps by identifying high-demand employment sectors and analyzing the skills needed to thrive in these industries (Alam et al., 2024a, 2024b).

This research is highly relevant to multiple stakeholders. It offers actionable insights for graduates seeking to enhance their employability, provides educational specialists and policymakers with evidence-based recommendations for curriculum reform, and guides government officials in fostering industry-academia collaboration. By addressing the skills gap, the study aims to bridge the divide between academic preparation and industry requirements, contributing to sustainable economic

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development in Bangladesh. To achieve its objectives, this study seeks to answer the following questions:

RQ1: What are the emerging sustainable employment areas during the 4IR in Bangladesh?

RQ2: To what extent do graduates of Bangladesh possess a skills gap for their future employability?

2. Literature review

2.1. Fourth industrial revolution

The 4IR refers to the rise of "cyber-physical systems," providing new opportunities for people and robots (Davis, 2016). Schwab (2015) argues that we are on the brink of technological development that will undeniably and significantly influence several parts of our lives, such as our way of living, jobs, and relationships. The transformation's scale, scope, and complexity will surpass any prior experiences in human history. The global restructuring of industries during the 4IR evokes a balance of excitement and concern. According to Jee (2017), the 4IR is anticipated to provide significant consumer advantages. Schwab (2015) predicts that implementing technological advancements will lead to a supply-side miracle, marked by substantial enhancements in productivity and effectiveness over a long period.

2.2. Industry 4.0 and the future employable opportunities

Distinguished academics argue that the 4IR will significantly impact the course of civilization, especially concerning trade, commerce, and government. People have little control over technology or the other changes the 4IR has brought forth. However, predicting the prospective opportunities linked to the 4IR is feasible. These factors encompass reduced barriers between innovators and markets, greater integration of artificial intelligence (AI), the fusion of various methods and fields, improved quality of life through robotics, and the interconnectivity of our lives facilitated by the Internet (Xu et al., 2018).

McKinsey & Company's study reveals that over 50 percent of present labor functions can be automated with current technologies (Xu et al., 2018). This automation has the potential to lead to substantial cost reductions for companies and the introduction of new and unique job prospects. Holler et al. (2014) state that the Internet of Things (IoT) may provide improved connections for devices, systems, and services that go beyond machine-to-machine (M2M) interaction. This relationship involves a diverse array of protocols, sectors, and activities. These devices are expected to enhance automation in many sectors and enable the adoption of advanced technologies, such as a smart grid.

2.3. Graduates skills gap during the 4IR

Over the last decade, the economy of Bangladesh has seen a continuous and extended era of economic development, with a gross domestic product (GDP) growth rate of over six percent yearly (Hussain & Haque, 2016). Research conducted by the Ministry of Finance has revealed a significant disparity between the ability to find employment and the limited access to opportunities for skill development. This mismatch is further exacerbated by the low skill level of workers, resulting in decreased efficiency and reduced incomes for both formal and informal sector employees. The country's growth in the long run is highly dependent on the prioritization of improving workers' skills. According to the Ministry of Finance (2018), there is a shortage of trained workers in many new disciplines within the information technology industry, including Robotics, Artificial Intelligence, and the Internet of Things (IoT). The main obstacles to adopting new technologies in Bangladesh are inadequate infrastructure, limited comprehension, availability of inexpensive labor, disagreement, insufficient government assistance, restricted ability to establish crucial systems, leadership preparedness, and scarcity of skilled workforce. Bangladesh faces challenges in adopting automation and digitization technologies due to insufficient infrastructure, lack of awareness, abundant cheap labor, opposition, limited government support, limited capacity to establish essential systems, leadership preparedness, and shortage of skilled workforce.

Fig. 1 illustrates the shortage of skilled human resources across technological firms in Bangladesh. There is still a discrepancy between the anticipated academic education level and software engineering proficiency. The employment market is experiencing a significant shortage of IoT and Software Testing professionals. While most professions often do not have difficulties filling lower and mid-level roles, they often face obstacles in identifying appropriate applicants for higher-level positions. Around 40 percent of organizations indicated a significant shortage of skills among their senior-level employees, particularly in the areas of application development (38 %), system administration (37 %), and data science (39 %). Over 30 % of firms have identified a substantial skills gap among mid-level professionals in different positions. The Ministry of Finance (2018) said that applications developer, web developer, database designer, system administrator, and Data Scientist jobs accounted for 34 %, 32 %, 33 %, 26 %, and 37 %, respectively. Industries are now experiencing a substantial transition and progress toward complete digitization and intelligent production processes to guarantee maximum efficiency. In this context, the next workforce must acquire new skills in automation, digitalization, and information technology to adjust to changing conditions efficiently (see Fig. 2).

3. Research method

3.1. Research design

This study employed a mixed-methods research technique with an inductive approach to investigate the potential for future employment during the 4IR and its significance in Bangladesh. The research utilized a concurrent design, wherein qualitative and quantitative data were collected simultaneously to understand the skills gap comprehensively.

Qualitative data were gathered based on the theoretical framework to explore emerging employment opportunities and the required skills in the 4IR. Semi-structured interviews were conducted with participants selected through purposive sampling. These interviews were recorded, transcribed, and analyzed using thematic coding to identify recurring patterns and themes.

Quantitative data were collected using survey research methods, which included semi-structured and structured questionnaires targeting university graduates and stakeholders from the Bangladesh Skills Development Program. The questionnaire was designed with a 5-point Likert Scale to measure perceptions and skill levels. The sample size of 1320 respondents was determined to ensure representation from both demand and supply sides, using random sampling to minimize bias.

Integrating qualitative and quantitative methods was crucial for

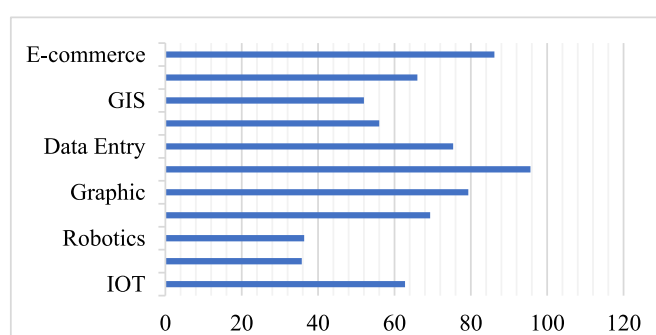

Fig. 1. Skilled human resources across technological firms in Bangladesh
Source: The ministry of finance (2018).

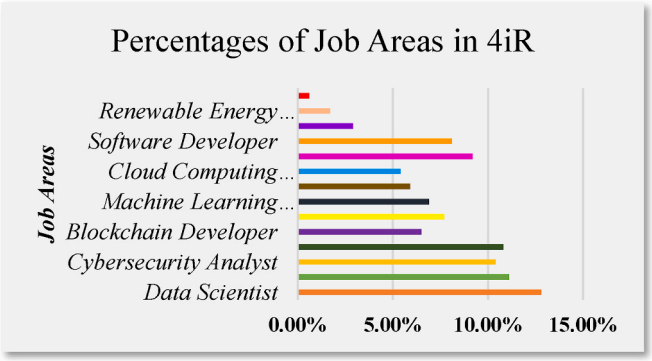

Fig. 2. The Areas for Future Employability Considering 4ir in Bangladesh
Source: Created by the authors.

cross-validation and enhancing the robustness of the findings. SPSS was used to analyze quantitative data, employing descriptive statistics to explain and interpret the results. Ethical considerations, including informed consent, data confidentiality, and secure information storage, were carefully addressed, ensuring the research process's integrity.

3.2. Detailed sampling

In this study, the researchers employed a multi-stage approach combining random purposive and snowball sampling to ensure a representative and comprehensive dataset.

For university graduates, a random purposive sampling methodology was utilized. This method allowed researchers to randomly select participants while considering their discernment and relevant skills to effectively address the study's objectives. This strategy balanced the need for diverse representation with the practicality of resource constraints. Ethical considerations were integral to this process, ensuring informed consent, participant privacy, and secure data handling to prevent potential harm.

For government policymakers, snowball sampling was adopted to identify and recruit participants. This approach involved an iterative process where initial contacts, carefully selected for their expertise and relevance, referred additional participants. To minimize bias, researchers meticulously verified the suitability of each participant before inclusion. This strategy ensured access to key informants who could provide valuable insights into policy-related aspects of skills development in the context of the 4IR. By combining these sampling methods, the study achieved a balanced representation of perspectives from both the supply and demand sides, ensuring the findings' reliability and depth.

3.3. Sample size analysis

Participants for this study were selected through a combination of random purposive and snowball sampling methods, ensuring diverse representation across qualitative and quantitative data collection. The qualitative sample consisted of 32 participants, including respondents from in-depth interviews and focus group discussions (FGDs). For the quantitative analysis, 1320 participants were surveyed using structured questionnaires.

The sample size of 1320 for the quantitative study was determined based on practical considerations, budgetary limitations, and statistical precision. The following formula derived from Taherdoost (2017) was utilized to determine the sample size:

$$n = \frac{Z^2 \times p(1-p)}{\varepsilon^2}$$

The sample size was determined assuming a 99.9 % confidence level ($Z = 3.29$) and a margin of error of 4.6 % ($\varepsilon = 0.046$). The estimated proportion of the population with the characteristics of interest (p) was conservatively set at 50 %. Using this formula, the calculated sample size was 1279. However, to facilitate balanced representation across all eight administrative divisions and ensure an even distribution of samples from various regions and universities, the total sample size was rounded up to 1320. This adjustment allowed for additional participants to be included while maintaining the sampling process's integrity.

Table 1 provides details of the sample distribution, with the number of entities (universities), unit sample sizes, and total sample sizes clearly outlined. This stratified approach aimed to balance data collection and minimize selection bias by considering student populations, institutional diversity, and regional representation.

3.4. Source of data

This study used a mix of questionnaires, focus groups, and inspections to collect information. Every discussion has been recorded, converted into written form, and examined systematically to identify recurring patterns and themes. The primary data sources for this research consist of government stakeholders and education professionals. The process of triangulating data from surveys, interviews, and focus groups enhances the validity of the results by including diverse viewpoints and facilitating the verification of information. This enhances the dependability and accuracy of the data sources.

3.5. Data analysis mechanism

The research used thematic analysis to examine the qualitative primary data collected via interviews and focus group discussions (FGDs). The research uses thematic analysis since it is a valuable tool for methodically characterizing phenomena. The versatility of this analysis method is generally acknowledged across several disciplines. The study used the recommendations of Braun and Clarke (2006) to perform thematic analysis.

The inquiry used the recorded responses and written records to deduce its findings. The methodology used in this study included field interviews, observations, and assessments. The data was methodically organized into subjects and sub-themes utilizing logical approaches. This was followed by a comparative analysis to derive findings. The study comprehensively analyzed the data collected for this research, including many criteria such as ideology, topics, participant features, and coding.

3.6. Theoretical framework

This study will use the Human Capital Theory since it aligns well with the research objective. Various studies have shown the importance of human capital development in terms of future employability, the

Table 1
Details sampling, locations, and specific size.

Location (Division)	No. of Entity (University)	Unit Sample Size	Total Sample
Dhaka	4	80	320
Chittagong	4	80	320
Rajshahi	3	60	180
Khulna	3	60	180
Barishal	2	40	80
Sylhet	2	40	80
Mymensingh	2	40	80
Rangpur	2	40	80
Total	22		1320

Note. The sample was estimated considering the total number of students, diversity, and institution type.
Source: Created by the authors.

emergence of skills gaps, and the transformation of skills (Hirsch-Kreinsen & Ittermann, 2021). The digital processes of globalization, driven by technological and scientific progress, inherently transform human civilization (Makarova et al., 2019).

In the context of the 4IR, sustainable employment relies on technically competent individuals who serve as the main drivers of production. Their expertise in information technology (IT) and their proficiency in related skills are crucial in driving economic activity (Lewandowski, 2018). A crucial factor is the advancement of technological innovations, global collaboration for progress, and institutional transformation, explicitly establishing a legal framework that can adapt to changing environments and is associated with contemporary specialized industries and AI-driven manufacturing processes (Zaborovskaia et al., 2020). Human capital remains crucial to promoting the economic growth of the IT-based industry (Pelinescu, 2015). In Bangladesh, the term "4IR" refers to the government's focus on implementing sustainable employment policies to enhance the country's human capital. Contemporary methodologies and tools for formulating plans must include the impact of ongoing technology advancements and the adaptability of the workforce (Warner & Wäger, 2019).

4. Findings

4.1. Sustainable employability areas during the 4IR

Participants were surveyed to explore potential employment opportunities in Bangladesh in the 4IR context. The findings highlight the growing importance of technology-driven sectors, with distinct trends emerging from quantitative and qualitative analyses. Based on data from 1320 respondents using a 5-point Likert scale (1 = Very Low Demand, 5 = Very High Demand), the study identified Artificial Intelligence (AI) as the most promising career field, with 553 participants rating it as having "High" and 266 respondents rated it as "Very High" demand. The mean score for AI was 3.63 (SD = 191.61), reflecting strong agreement on its significance. Data science emerged as the second most desirable field, with 462 respondents perceiving it as "High" and 198 respondents seeing it as "Very High" in demand, resulting in a mean score of 3.30 (SD = 132) [see Table 2].

Digital marketing ranked third, with 310 participants rating it as "High" and 209 rating it "Very High" in demand (M = 3.01, SD = 57.75). Other fields, such as e-commerce, cybersecurity, and robotics, were rated moderately to highly significant, with average mean scores ranging from 2.95 to 2.63. These findings underscore the increasing prominence of technical skills across various domains, driven by the rapid transformation of industries during the 4IR.

Qualitative insights from semi-structured interviews and focus group discussions supported these quantitative results. One respondent remarked,

> *"Traditional jobs are being altered to incorporate new industries like cybersecurity, digital advertising, and robotics, as a result of the 4IR shaping the business landscape"* (Res 15).

Another participant emphasized the growing role of data-driven decision-making, stating,

> *"The field of data science and analytics is expected to experience substantial expansion as businesses utilize data to make knowledgeable choices"* (Res 17).

Additionally, the rising dominance of online platforms was highlighted, with one participant noting,

> *"The need for professionals proficient in e-commerce and online marketing will grow as internet platforms keep gaining control over consumer engagements"* (Res 09).

The findings suggest that the demand for emerging skills varies across sectors, regions, and demographics, reflecting the diverse nature of Bangladesh's labor market. These results highlight the critical need for tailored skill development initiatives and targeted policy interventions to ensure that graduates can meet the evolving demands of the 4IR-driven economy.

4.2. Skills gap and mismatch among graduates

Table 3 displays the proportion of the difference in skills between those obtained and those necessary for work in the 4IR. The data indicates that 45.2 % of the participants indicated a considerable disparity between their abilities and the skills necessary for work in the 4IR. Nevertheless. 28.6 % of respondents expressed dissatisfaction with the unavailability of the gap.

One respondent added,

> *' …. a noticeable gap, especially in the successful use of advancing technologies. The information I gained from my studies provided the foundation, but active participation was crucial during the time of the 4IR.'* (Res 01)

Another respondent mentioned,

> *'Recognizing the deficiency in abilities is the initial phase. I firmly advocate actively pursuing chances to improve my skills and staying up-to-date with the latest developments in the field to close this distance effectively.'* (Res 13)

Table 4 illustrates the proportion of skill mismatch between the abilities possessed by graduates and the skills required by jobs in the context of the 4IR. The findings indicate that a combined 47.7 % of respondents acknowledged a significant disparity between the abilities possessed by graduates and the skills demanded for jobs in the 4IR. They

Table 3
The Skills gap between the skills acquired and the skills required for employment in 4iR.

	Frequency	Percent	Valid Percent	Cumulative Percent
1	377	28.6	28.6	28.6
2	94	7.1	7.1	35.7
3	95	7.2	7.2	42.9
4	596	45.2	45.2	88.0
5	158	12.0	12.0	100.0
Total	**1320**	**100.0**	**100.0**	

Source: Created by the authors.

Table 2
Analysis of employment area based on student's perceptions.

Employment area	Very low	Low	Moderate	High	Very high	Standard deviation	Mean	Total
AI	66	118	317	553	266	191.6077	3.63	1320
Data Science	132	198	330	462	198	132	3.30	
E-commerce management	198	264	396	330	132	104.3552	2.95	
Cybersecurity	293	322	287	282	136	73.21544	2.73	
Robotics	316	341	293	192	178	74.25295	2.68	
Digital Marketing	249	214	338	310	209	57.75379	3.01	

Source: Created by the authors.

Table 4
The mismatch between graduate skills and job requirements in the 4IR.

	Frequency	Percent	Valid Percent	Cumulative Percent
1	188	14.2	14.2	14.2
2	189	14.3	14.3	28.6
3	314	23.8	23.8	52.3
4	377	28.6	28.6	80.9
5	252	19.1	19.1	100.0
Total	**1320**	**100.0**	**100.0**	

Source: Created by the authors.

believe that the mismatch arises because of a deficiency in information. However, 28.5 % of the participants responded negatively (Highly disagree and disagree), indicating a skill mismatch between the talents possessed by graduates and the need for jobs in the 4IR.

One respondent added,

'Though my schooling provided me with essential information, it is necessary to prioritize practical experiences and teamwork to tackle the skills shortage effectively.' (Res 29)

Another said,

'Inadequate skill level, particularly in advanced data interpretation and coding domains. Closing this divide necessitates a dedication to continuous education and advancement in one's career.' (Res 14)

5. Discussion

5.1. Sustainable employable opportunities during the 4IR

The 4IR presents significant opportunities for increased employability and productivity within the work market. The advent of the 4IR has the potential to accelerate the creation of new job prospects in expanding sectors that see enhanced levels of efficiency and productivity (Fox & Signé, 2021). The research indicates that data scientists are the most favorable industry in Bangladesh's context of the 4IR. Osunsanmi et al. (2023) reported the same results. Experts in the real estate field will effectively analyze property investment data using specialized technologies, leading to a successful investigation. Rapanyane and Sethole (2020) argued that globalization is causing significant changes worldwide, affecting businesses, governments, and people. Advanced technologies, such as artificial intelligence and computerization, are causing a transformation in the lifestyle and work patterns of individuals in South Africa. Bank (2018) said electronic commerce in the context of the 4IR has significant potential for promoting economic growth. However, it also presents challenges due to several elements, such as economic, legal, structural, and social issues.

The results indicated that although most students were familiar with the concept of the 4IR, their comprehension of digital literacy and its fundamental components was pretty restricted. The inquiry proposed conducting a study that would specifically target recent college graduates to assess their proficiency in computer skills, understanding of concepts related to the 4IR, and preparedness for further education (Sithembiso, 2020). Students' preparedness for 4IR technologies is greatly influenced by their attributes, comprehension of 4IR technologies, and managerial factors. To promote the adoption of the 4IR, it is crucial for higher education institutions to give priority to educational initiatives, the development of skills, and the provision of technology resources (Al-Maskari et al., 2022; Islam & Alam, 2023). The students possess knowledge of Industry 4.0 but need more guidance on its implementation (Ujakpa et al., 2020). Reducing the discrepancy in students' understanding of patterns connected to the 4IR and the curricular content is very important (Cele et al., 2023).

5.2. Skills mismatch of university graduates of Bangladesh

Based on the study, most participants said there is a discrepancy between their abilities and the skills needed for work in the 4IR. The 4IR has been catalyzed by digitalization, artificial intelligence (AI), and the uncertainty surrounding job paths. As a result, this revolution has caused a shortage of skills, necessitating retraining and increased unpredictability in job opportunities. Skill obsolescence happens when there is a discrepancy between work requirements and an individual's talents, as Abe et al. (2021) predicted. Professionals have a decent understanding but lack significant proficiency in data analytics, cyber security, and interactions between humans and machines (Adepoju & Aigbavboa, 2020a). University students and new graduates in Bangladesh lack interpersonal communication abilities, English fluency, time management skills, and problem-solving capacity. A skills gap presents difficulties for the nation's labor market as it endeavours to harness the potential of its youthful population and adapt to the 4IR (Khatun & Nawrin, 2021; Alam et al., 2024a, 2024b; Alam & Ogawa, 2024). It is advisable to implement curricular reform to promote human capital development (Adepoju & Aigbavboa, 2020b). They believe that educational institutions should prioritize efforts to reduce the current disparity. Otherwise, the disparity in skills will persist. The study proposes a new approach for business schools to develop ethical leaders for the post-COVID-19 workplace and the 4IR. The approach aims to enhance professional competence and contribute to positive societal change in line with the United Nations' Sustainable Development Goals (Diaz & Halkias, 2021).

Reskilling is creating a strategy to help people gain new skills and prepare them to take on new responsibilities to meet the requirements of their profession, industry, and evolving technology (Partovi, 2018; Alam et al., 2024a, 2024b). Students must get proper training to address the difficulties posed by the 4IR. Workers need adequate guidance to adapt and modify themselves following these alterations (Stofile et al., 2023). Adapting to the new working practices presented by the 4IR necessitates possessing leadership skills. Competent leadership in the 4IR necessitates a range of essential leadership skills, including adaptability, astute interpersonal relationships, advanced understanding, effective information management and analysis, and the practical application of knowledge across different domains (Alade & Windapo, 2020).

6. Conclusion

The Fourth Industrial Revolution (4IR) propels worldwide change via technological innovation to optimize labor processes and enhance productivity. Artificial intelligence (AI) can replace human laborers, enhancing productivity and unemployment. Nevertheless, it is anticipated that technological progress will improve work prospects by fostering the enhancement of abilities. Bangladesh is progressively transitioning towards embracing the 4IR. Nevertheless, there are other obstacles to overcome, such as inadequate technical infrastructure, slow internet connection, and a discrepancy between the skills possessed by graduates and those required by the job market. The findings revealed a multitude of employment prospects in the realm of modern technology for those who have completed their university education. The grads are also interested in careers as data scientists, Artificial Intelligence Engineers, cybersecurity analysts, robotics engineers, software developers, and digital marketing strategists. However, the data revealed a significant disparity and mismatch in capabilities among university graduates in Bangladesh. Therefore, educational institutions should prioritize measures aimed at diminishing the existing imbalance. Furthermore, it is essential to implement skilling and upskilling programs in companies. Leadership, time management, and critical thinking skills are essential in the context of the 4IR.

This research aimed to provide an accurate and comprehensive analysis of the 4IR and the discrepancy in capabilities among recent graduates. Nevertheless, the investigation is subject to time and

monetary constraints. Additionally, there has been a concern about gender parity throughout the sample collection process since the proportion of female learners in the skills development process is much lower than that of male learners (Alam et al., 2024a, 2024b). Therefore, this study opens opportunities for future research, such as conducting a comparative analysis between developed nations and Bangladesh to assess the effectiveness of 4IR preparedness, learning from developed nations on strategies to address skills mismatch, sustainable employment, and enhancing graduates' readiness.

7. Recommendations

Based on the study, some recommendations are given as follows.

7.1. Curriculum development

To enhance the progress of advanced technology education and scholarly pursuits, it is necessary to thoroughly evaluate the technological infrastructure of educational institutions and provide suggestions for investments in hardware, software, and internet connectivity. Collaboration between graduates, educational authorities, and government officials is essential to promote awareness and advancement of technology capabilities.

7.2. Industry-academia collaboration

The curriculum of technical universities should be revised and enhanced. Collaborative endeavours are necessary among many stakeholders to regularly evaluate and improve curricula, ensuring they align with the requirements of the technology sector in the 4IR.

7.3. Collaborative actions among graduates and the government authorities

Collaboration among government, education, and business is necessary to provide a comprehensive structure that improves the technological skills of graduates, enabling them to be prepared for the 4IR. The institutions should organize career counseling sessions, seminars, and workshops with the assistance of the industries to provide graduates with the necessary abilities.

Ethics guidelines and approval statement

This research manuscript is the author's original work, which has not been submitted elsewhere for publication. This study involved human participants for survey purposes, and all procedures were conducted following ethical standards. Department of Japanese Studies, 2023-ECA-STD-3/89409, November 15, 2023.

Declaration of the use of AI

This manuscript did not use any AI-generated content.

The standard conditions of this approval are

(a) Conduct the project strictly according to the proposal submitted and grant ethics approval, including any amendments required by the review board.
(b) Please advise the review board immediately of any complaints or other issues related to the project that may warrant a review of the ethical approval of the research project.
(c) Make a submission for approval of amendments to the approved project before implementing such changes.
(d) Advise in writing if the project has been discontinued.

CRediT authorship contribution statement

Md Jahangir Alam: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Shiblee Noman:** Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md Nurul Islam Mujib:** Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Wasik Sajid Khan:** Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Formal analysis, Conceptualization.

Availability of data and materials

To protect and preserve respondent confidentiality, the data sets developed or analyzed during the current study are inaccessible to the public. Still, they are available upon reasonable request from the corresponding author.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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